

Roll No. 0941601511
REF NO. Conf/Even/N/2025-26/21

B. Sc. (Hons) Semester IV Examination
2025 - 26 (NEP)

BOTANY

Paper No. BOTMJ45 : Plant Ecology

Time : 2½ hours

Full Marks : 70

**(Write your Roll No. at the top immediately
on the receipt of this question paper)**

The figures in the right-hand margin indicate marks

Answer **five** questions including Question 1
which is compulsory

1. Describe briefly/define/answer, in
about 50 words each of the following : 7×2

(a) Synthetic community characters

(b) Soil forming factors

(c) Define phenology.

(d) Raunkiaer's life forms

(e) Distinguish between phototrophs
and chemotrophs

(2)

(f) Function of stomatal hairs in plant adaptation

(g) What is ecesis?

2. Describe soil structure and explain soil profile with the help of a suitable diagram. 14

3. Write short notes on *any two* of the following : 2×7

(a) Atmosphere

(b) Adaptations in hydrophytes

(c) Role of temperature in distribution of vegetation.

4. Why biotic interaction is important in community? Describe various types of positive and negative biotic interactions in a community. 14

5. Write short notes on *any two* of the following : 2×7

(a) Survivorship curve

(b) Biological spectrum

(c) Causes of ecological succession.

(3)

6. Define community. Discuss various quantitative characters of the community.

14

7. Write short notes on *any two* of the following :

2×7

(a) Nitrogen cycle

(b) Models of energy flow in the ecosystem

(c) Ecological pyramids.

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